name: ______

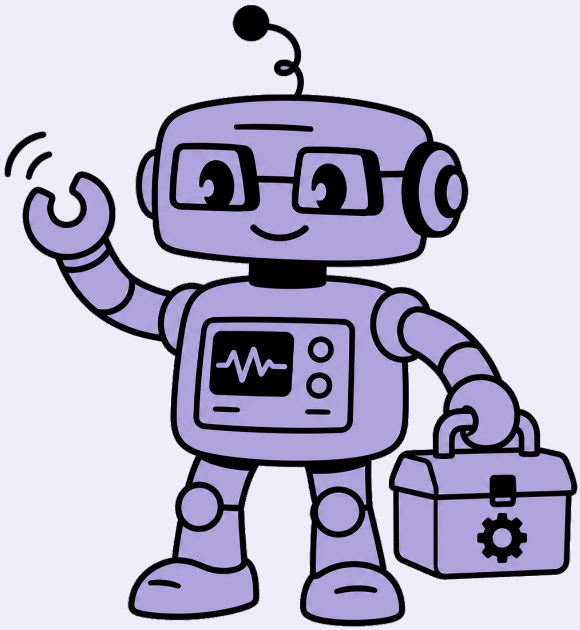
openhouse

robotics · level 2

electronics

your experience & progress book

ages 5-8



the four skills

the skills you grow

in every machine you grow the same four skills. look for the little icon at the top of each page – it tells you which skill you're using.

building & making
you build a real machine from blocks and circuit cards, step by step.

observing & understanding
you watch what your circuit does – then work out *why* it works.

problem solving
you think up and design your own ways to make it better.

presenting
you present your machine to the group, walk through how it works, and answer their questions.

what's inside

table of contents

each machine takes you through all four skills – build it, watch it, understand it, improve it, and present it.

before you begin

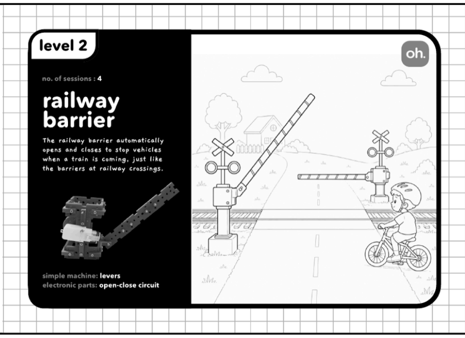
staying safe with electricity

electricity is a helper, but we treat it with care. four rules for every engineer:

- the battery blocks in your kit are **safe** to touch and build with – never use wall plugs or house electricity.
- never put wires or blocks in your mouth, or near water.
- if a wire feels warm, switch it off and tell your teacher.
- always build with your teacher nearby – engineers work safely.

for the child

meet the railway barrier



your railway barrier – it lifts and lowers when you flip the switch.

Every morning, Aarav's school bus stops at the railway crossing. A striped barrier comes down, a bell rings, and everyone waits as the train rushes past. Then the barrier lifts and the bus rolls on. how does it know when to go up and down? today, think like an engineer.

think about it

have you ever waited at a railway crossing? what made the barrier go up, then come down? draw it, or tell a friend.

words to know

circuit · switch · electricity

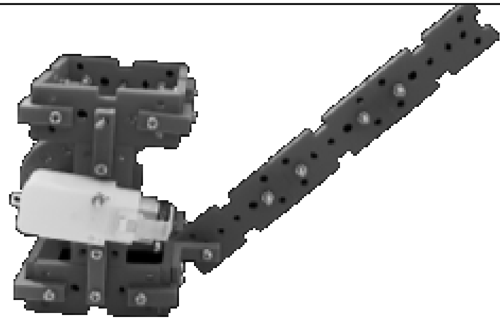
did you know?

the **electricity** flows round the loop to make it work – but only when the loop is joined all the way round.

👩 for the child

🔧 building & making

build your railway barrier.



this is what you're building.

📖 **use your model manual**

follow your **railway barrier manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: add the barrier arm and the switch, so it **lifts and lowers when you flip the switch**.

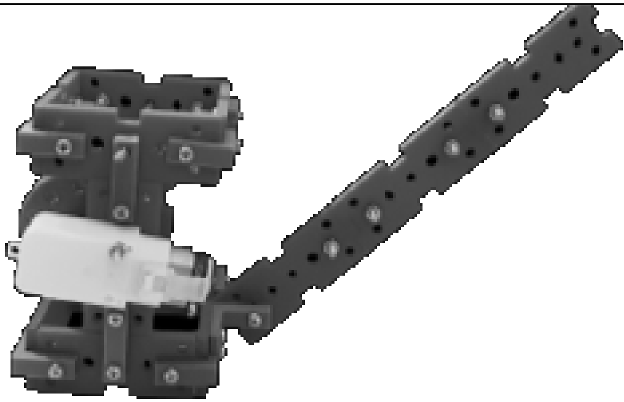
railway barrier

🔧 building & making

for the child

⭐ name the parts

✎ on the picture of your model, circle each part and write its name next to it.



label these four parts

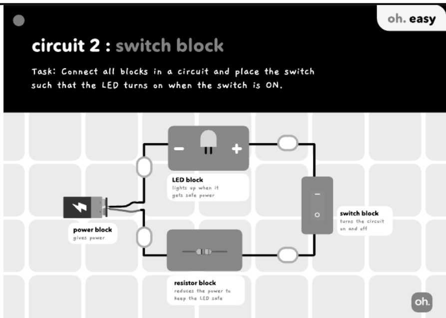
the motor | the battery | the switch | the barrier arm

⭐ **bonus:** draw an arrow to show how it moves.

for the child

observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when you pressed the switch?

◯ the light came on when the switch was on, and went off when it was off

◯ the light kept blinking by itself

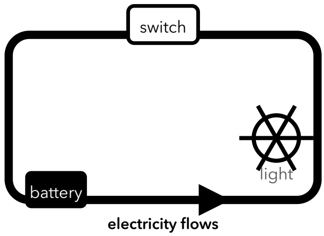
◯ the light stayed off the whole time

✎ show your teacher what happened. did it work? ☐ **yes** ☐ **not yet – try again together**

🧑 for the child

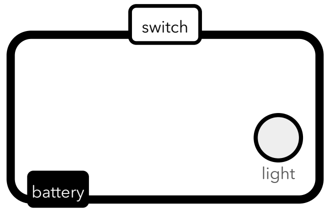
🔬 observing & understanding

the big idea – why does it work?



the loop is joined

☐



the loop has a gap

☐

the switch opens and closes the loop. when the loop is **joined**, the **electricity** flows all the way round (see the arrows) and it works. when there's a **gap**, nothing flows.

✎ can you finish it?

for the barrier to move, the loop must be…

- ◯ joined all the way round
- ◯ broken with a gap in it

for the child

problem solving

how could you make it even better?

✎ circle the idea that works best.

1 · make the crossing easy to see at night

so drivers slow down in the dark. which idea works best?

- ◯ add a bright light to the barrier
- ◯ paint it a dull grey
- ◯ make it thinner

2 · warn people before the barrier closes

how could it tell them to step back?

- ◯ make it beep and flash first
- ◯ close as fast as it can
- ◯ give no warning

3 · keep the control box working in the rain

what would protect it best?

- ◯ put a little roof over the box
- ◯ pour water on the wires
- ◯ leave it out in the rain

⭐ **your idea:** think of one more way to make it better. draw it and label it.

✎ draw your idea here

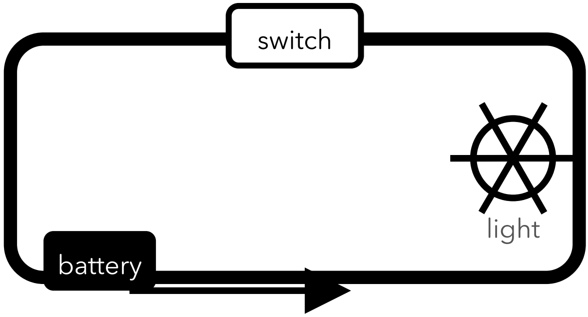
for the child

problem solving

engineer challenge – design your own

you want a **light that switches on when the barrier closes**, to warn cars.

here's an example – a switch turns a light on and off:



one complete loop

✎ **tick the blocks you would need**

- power block
- switch block
- light block
- motor block
- wires

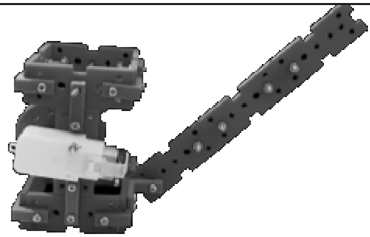
✎ **now draw your own circuit** – join the blocks in a **complete loop** so it works.

✎ draw your circuit here

for the child

presenting

be the engineer – present your machine



show your railway barrier to the group and take them through it:

1 **present** – show your railway barrier to the group – say what it does and point to each block.

2 **walk through** – show how it works – trace how the electricity goes round the loop.

3 **say why** – say why one block matters – what happens if you take it out?

4 **help a friend** ⭐ – answer one question about it – or help a friend fix theirs.

at-home show & tell

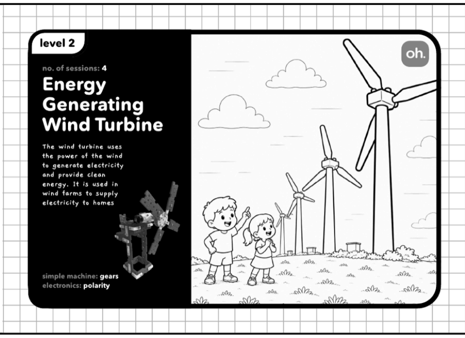
take it home and show a family member. explain: **what problem does it solve? how does it work?** (it keeps cars and people safe at the railway crossing.)

family member: ____________________

signature: ____________________

for the child

meet the wind turbine



your wind turbine – it spins in the wind and lights up an led.

On a windy hill, Meera watches a giant white windmill turn, slow and steady. Her teacher says it isn't grinding grain – it's making electricity for the whole village. Meera wonders: how can spinning blades light up a bulb? today, you'll find out.

think about it

where have you seen something spin in the wind? what do you think that spinning could power?

words to know

energy · led · generator

did you know?

a spinning motor can **make** electricity – then it's called a **generator**.

wind turbine

🔧 building & making

build your wind turbine.

this is what you're building.

📖 **use your model manual**

follow your **wind turbine manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: connect the motor to the led block, so it **spins in the wind and lights up an led**.

wind turbine

🔧 building & making

for the child

⭐ name the parts

✎ on the picture of your model, circle each part and write its name next to it.

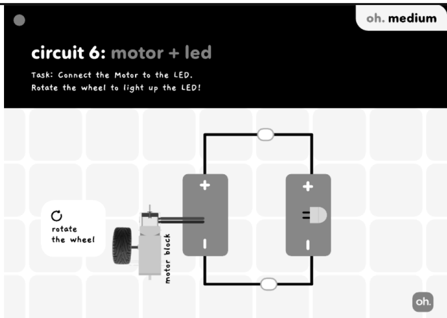
label these four parts

the blades | the motor | the led | the wires

⭐ **bonus:** draw an arrow to show how it moves.

observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when the blades spun faster?

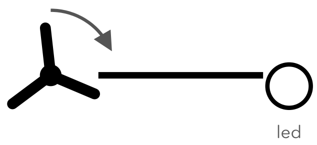
- ◯ the led glowed brighter
- ◯ the led went dark
- ◯ nothing changed at all

✎ show your teacher what happened. did it work? ☐ **yes** ☐ **not yet – try again together**

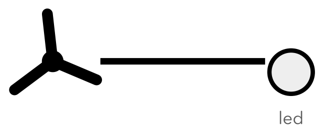
for the child

🔬 observing & understanding

the big idea – why does it work?



spinning fast



barely spinning

the spinning blades turn the motor into a tiny **generator** – it makes electricity. the **faster** it spins, the more power, and the brighter the led. (an led only lights one way round.)

✎ can you finish it?

to make the led glow brighter, the blades should spin…

- ◯ faster
- ◯ slower

wind turbine

for the child

problem solving

how could you make it even better?

✎ circle the idea that works best.

1 · catch more wind so it spins faster

which idea would help most?

- ◯ make the blades bigger and turn them to face the wind
- ◯ make the blades tiny
- ◯ point them away from the wind

2 · light the led even in a gentle breeze

what could you try?

- ◯ spin it by hand to give it a start
- ◯ wait for a storm
- ◯ take the led out

3 · show how much power it makes

what would show it best?

- ◯ a bright led that glows more when it spins faster
- ◯ a sticker
- ◯ nothing

your idea: think of one more way to make it better. draw it and label it.

✎ draw your idea here

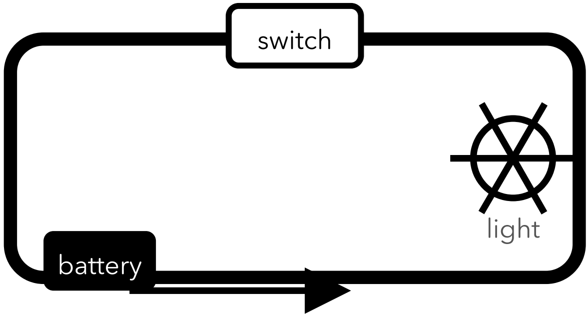
for the child

problem solving

engineer challenge – design your own

you want the **led to light up when the blades spin**.

here's an example – a switch turns a light on and off:



one complete loop

✎ **tick the blocks you would need**

- motor block
- led block
- resistor block
- wires
- switch block

✎ **now draw your own circuit** – join the blocks in a **complete loop** so it works.

✎ draw your circuit here

for the child

🗣️ presenting

be the engineer – present your machine

show your wind turbine to the group and take them through it:

1 **present** – show your wind turbine to the group – say what it does and point to each block.

2 **walk through** – show how it works – trace how the electricity goes round the loop.

3 **say why** – say why one block matters – what happens if you take it out?

4 **help a friend** ⭐ – answer one question about it – or help a friend fix theirs.

🏠 at-home show & tell

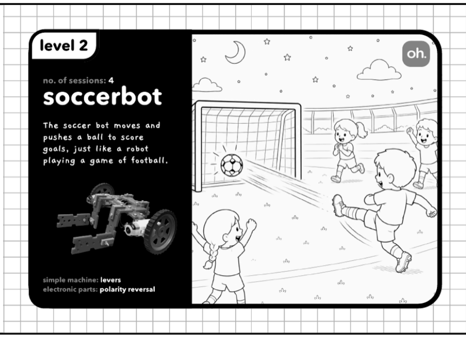
take it home and show a family member. explain: **what problem does it solve? how does it work?** (it turns wind into electricity to light a lamp.)

family member: ____________________

signature: ____________________

for the child

meet the soccer bot



your soccer bot – it drives forward and back when you flip the switch.

It's the big match! Kabir grips his controller as his little robot races across the floor, nudges the ball, and – GOAL! But to defend, he has to make it reverse, fast. how can one robot go both forwards and backwards? that's your challenge.

think about it

think of a toy that moves forward and back. how do you make it change direction?

words to know

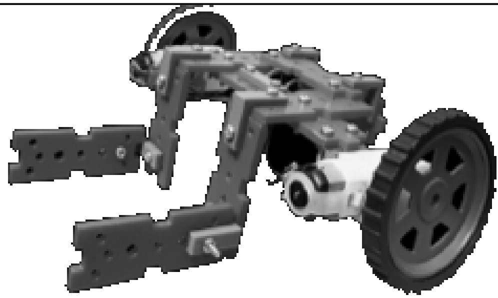
motor · direction · reverse

did you know?

swap the two wires on a motor and it spins the **other** way – that's **reversing**.

🔧 building & making

build your soccer bot.



this is what you're building.

📖 **use your model manual**

follow your **soccer bot manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: wire the motors through the dpdt switch, so it **drives forward and back when you flip the switch**.

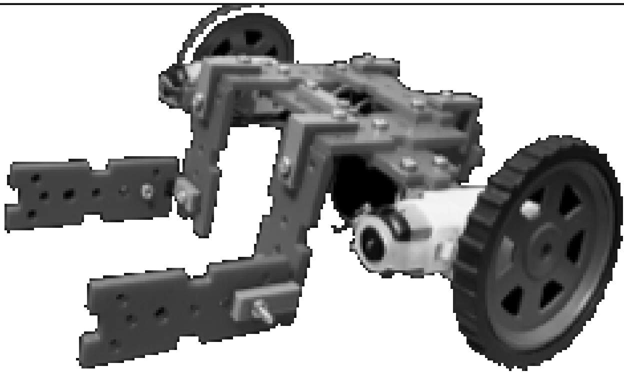
soccer bot

🔧 building & making

for the child

⭐ name the parts

✎ on the picture of your model, circle each part and write its name next to it.



label these four parts

the motor | the wheels | the battery | the switch

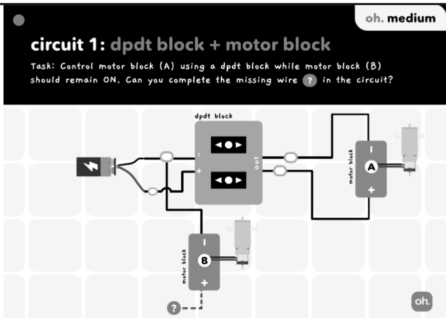
⭐ **bonus:** draw an arrow to show how it moves.

for the child

soccer bot

observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when you flipped the dpdt switch?

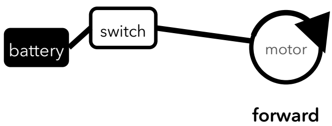
- ◯ the motor spun the other way, so the bot reversed
- ◯ the motor stopped forever
- ◯ the wheels fell off

✎ show your teacher what happened. did it work? ☐ **yes** ☐ **not yet – try again together**

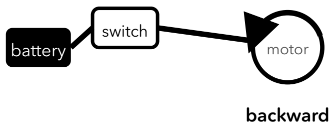
for the child

observing & understanding

the big idea – why does it work?



switch one way



switch the other way

flip the switch and the electricity flows the **other way** – so the motor spins backwards. that's how the same bot goes **forward and back**.

✎ can you finish it?

to make the bot reverse, you…

- ◯ flip the switch the other way
- ◯ add more wheels

for the child

soccer bot

problem solving

how could you make it even better?

✎ circle the idea that works best.

1 · make the bot turn to chase the ball

which idea would work?

- ◯ give each wheel its own motor and switch
- ◯ glue the wheels still
- ◯ remove a wheel

2 · make it quicker off the mark

what could help?

- ◯ a fresh, fully-charged battery
- ◯ a flat battery
- ◯ no battery

3 · stop it running off the pitch

what would help the player?

- ◯ a switch to turn it off fast
- ◯ no switch at all
- ◯ tape over the wheels

⭐ **your idea:** think of one more way to make it better. draw it and label it.

✎ draw your idea here

soccer bot

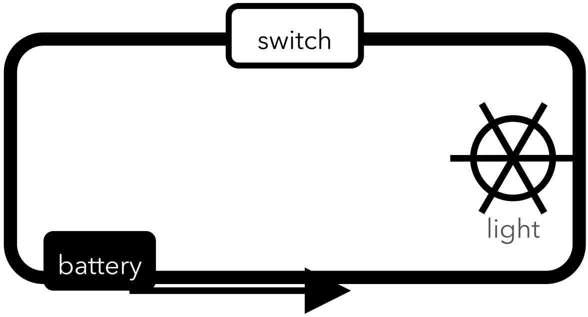
for the child

problem solving

engineer challenge – design your own

you want a **switch that makes the bot go forward and back.**

here's an example – a switch turns a light on and off:



one complete loop

✎ **tick the blocks you would need**

- power block
- dpdt switch
- motor block
- wheels
- wires

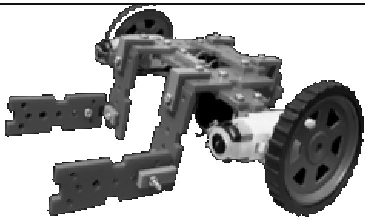
✎ **now draw your own circuit** – join the blocks in a **complete loop** so it works.

✎ draw your circuit here

for the child

presenting

be the engineer – present your machine



show your soccer bot to the group and take them through it:

1 **present** – show your soccer bot to the group – say what it does and point to each block.

2 **walk through** – show how it works – trace how the electricity goes round the loop.

3 **say why** – say why one block matters – what happens if you take it out?

4 **help a friend** ⭐ – answer one question about it – or help a friend fix theirs.

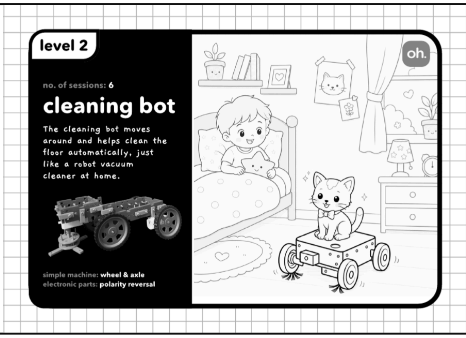
at-home show & tell

take it home and show a family member. explain: **what problem does it solve? how does it work?** (you drive it to chase the ball and score goals.)

family member: ______________________________

signature: ______________________________

meet the cleaning bot



your cleaning bot – it drives along and sweeps at the same time.

Diya's room is a mess again! She wishes for a little robot that could drive around AND sweep at the same time. Her uncle smiles: one battery can do more than one job at once. how? today, you'll build it and see.

think about it

if a robot could do two jobs at the same time, which two jobs would you give it?

words to know

power · parallel · current

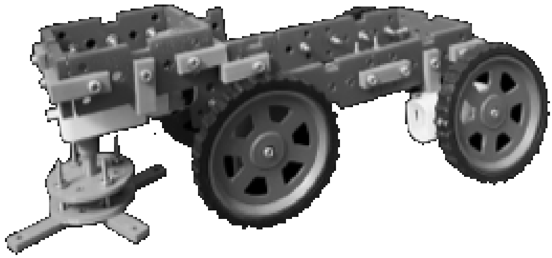
did you know?

when one battery runs two things at once, we say they are wired in **parallel**.

👩 for the child

🔧 building & making

build your cleaning bot.



this is what you're building.

📖 **use your model manual**

follow your **cleaning bot manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

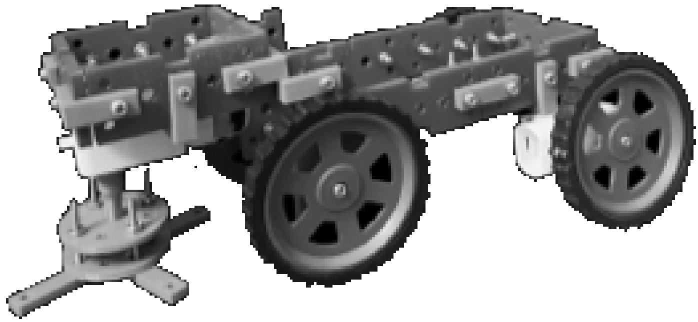
✎ next: wire both motors to the one battery, so it **drives along and sweeps at the same time**.

for the child

🔧 building & making

⭐ name the parts

✎ on the picture of your model, circle each part and write its name next to it.



label these four parts

the wheels' motor

the brush's motor

the battery

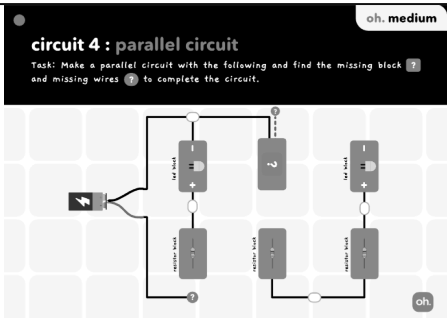
the brush

⭐ **bonus:** draw an arrow to show how it moves.

for the child

observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when you switched it on?

◯ the wheels AND the brush both worked together

◯ only one worked, never both

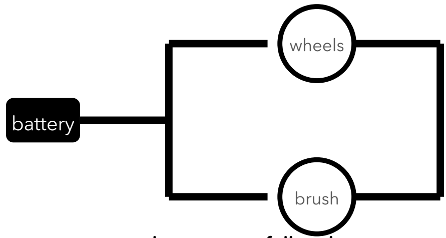
◯ nothing moved

✎ show your teacher what happened. did it work? ☐ **yes** ☐ **not yet – try again together**

for the child

🔬 observing & understanding

the big idea – why does it work?



one battery, two full paths

the power from **one battery** splits and runs **both** motors at once – each along its own path – so the bot can **drive and sweep** at the same time.

✎ can you finish it?

one battery in your bot can run…

- ◯ two jobs at the same time
- ◯ only ever one job

cleaning bot

for the child

problem solving

how could you make it even better?

✎ circle the idea that works best.

1 · reach dust in the corners

which idea would help?

◯ add a small brush that sticks out to the side

◯ make the bot wider

◯ remove the brush

2 · keep going for longer

what would help most?

◯ a fresh battery with more power

◯ a smaller battery

◯ no battery

3 · clean and drive without stopping

how does your bot already do this?

◯ one battery powers both jobs at once

◯ it does one job then the other

◯ it can only drive

⭐ **your idea:** think of one more way to make it better. draw it and label it.

✎ draw your idea here

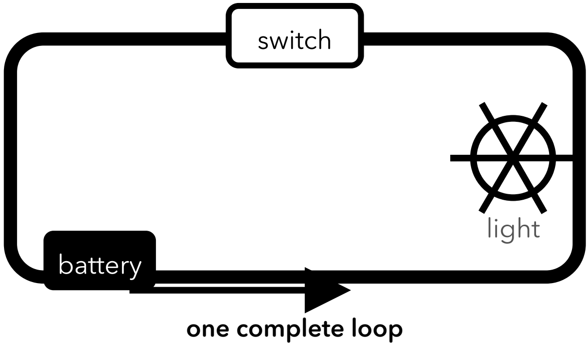
for the child

problem solving

engineer challenge – design your own

you want **one battery to run the wheels AND the brush** together.

here's an example – a switch turns a light on and off:



✎ **tick the blocks you would need**

- power block
- motor block (wheels)
- motor block (brush)
- wires
- switch block

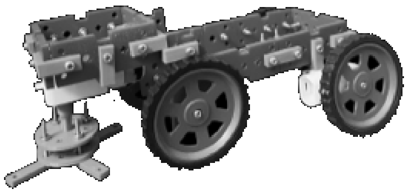
✎ **now draw your own circuit** – join the blocks in a **complete loop** so it works.

✎ draw your circuit here

for the child

presenting

be the engineer – present your machine



show your cleaning bot to the group and take them through it:

1. **present** – show your cleaning bot to the group – say what it does and point to each block.
2. **walk through** – show how it works – trace how the electricity goes round the loop.
3. **say why** – say why one block matters – what happens if you take it out?
4. **help a friend** ⭐ – answer one question about it – or help a friend fix theirs.

at-home show & tell

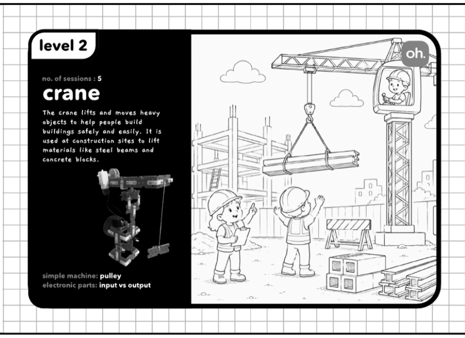
take it home and show a family member. explain: **what problem does it solve? how does it work?** (it sweeps the floor while it drives around.)

family member: ______________________________

signature: ______________________________

for the child

meet the sensor-controlled crane



your sensor-controlled crane – it lifts the load when the sensor is triggered.

At the port, huge cranes lift heavy boxes off the ships. Rehan notices something clever – the crane seems to KNOW when a box is under it, and only then does it lift. how can a machine sense and then act all by itself? let's build one.

💭 think about it

where have you seen a machine that notices something and then acts by itself – like a door that opens as you walk up?

words to know

sensor | input | output

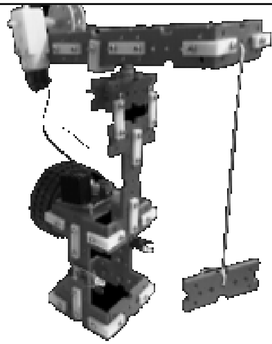
💡 did you know?

a machine part that **senses** the world (like light or an object) is called a **sensor**.

for the child

🔧 building & making

build your sensor-controlled crane.



this is what you're building.

📖 **use your model manual**

follow your **sensor-controlled crane manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: wire the sensor to the motor, so it **lifts the load when the sensor is triggered**.

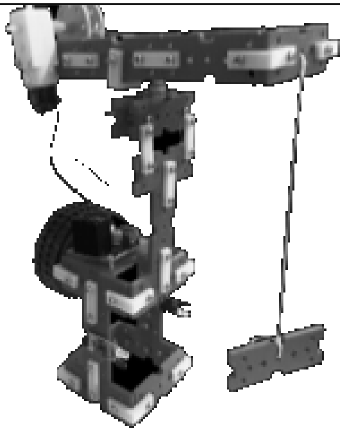
sensor-controlled crane

🔧 building & making

for the child

⭐ name the parts

✎ on the picture of your model, circle each part and write its name next to it.



label these four parts

the sensor | the motor | the hook | the battery

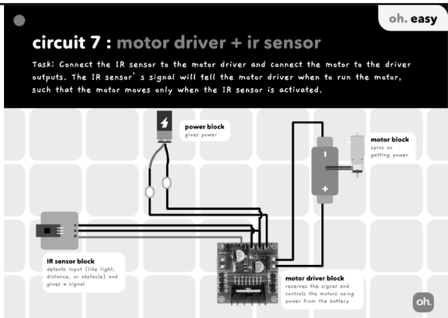
⭐ **bonus:** draw an arrow to show how it moves.

for the child

sensor-controlled crane

observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when something came near the sensor?

◯ the sensor noticed it and the motor lifted the load

◯ nothing happened at all

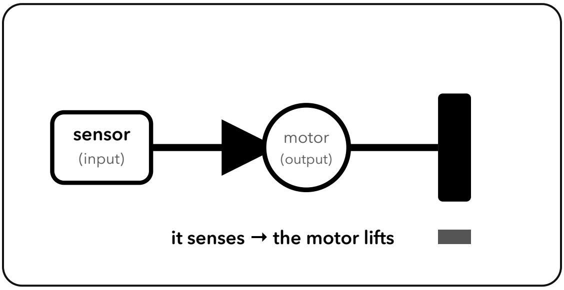
◯ the battery ran out

✎ show your teacher what happened. did it work? ☐ **yes** ☐ **not yet – try again together**

for the child

🔬 observing & understanding

the big idea – why does it work?



the **sensor** is the **input** – it senses when something is there. the **motor** is the **output** – it acts. the input tells the output what to do.

✎ can you finish it?

the sensor's job is to…

◯ sense, then tell the motor to act

◯ do nothing at all

for the child

problem solving

how could you make it even better?

✎ circle the idea that works best.

1 · lift a heavier box

which idea would help?

◯ use a pulley so it lifts with less effort

◯ use a thinner string

◯ take the motor off

2 · stop before it drops the load

what would help?

◯ a sensor that tells the motor when to stop

◯ no sensor

◯ close your eyes

3 · know when something is in the way

what does the sensor do?

◯ it senses the object and signals the motor

◯ it makes noise for fun

◯ nothing useful

⭐ **your idea:** think of one more way to make it better. draw it and label it.

✎ draw your idea here

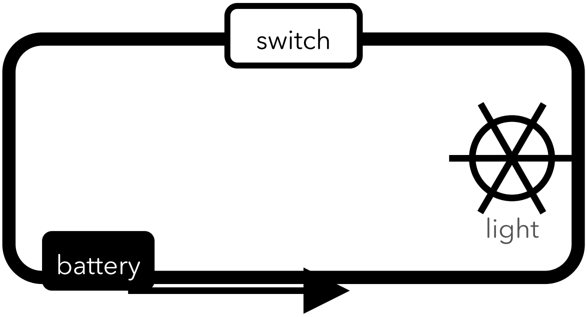
for the child

problem solving

engineer challenge – design your own

you want the **sensor to make the motor lift** when it senses a load.

here's an example – a switch turns a light on and off:



one complete loop

tick the blocks you would need

- power block
- sensor block
- motor block
- pulley & hook
- wires

now draw your own circuit – join the blocks in a **complete loop** so it works.

draw your circuit here

for the child

presenting

be the engineer – present your machine

show your sensor-controlled crane to the group and take them through it:

1 **present** – show your sensor-controlled crane to the group – say what it does and point to each block.

2 **walk through** – show how it works – trace how the electricity goes round the loop.

3 **say why** – say why one block matters – what happens if you take it out?

4 **help a friend** ⭐ – answer one question about it – or help a friend fix theirs.

at-home show & tell

take it home and show a family member. explain: **what problem does it solve? how does it work?** (it senses a load and lifts it, all by itself.)

family member: ____________________

signature: ____________________